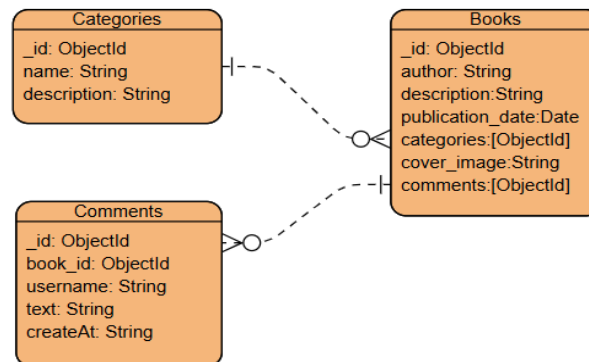


The **Book Management System** enables users to **view books, add new ones, and manage comments**. Built with **Node.js, Express.js, and MongoDB**, the backend provides a **RESTful API** for book and comment management. The **React.js frontend** consumes these APIs to dynamically display books and comments, ensuring a seamless user experience.

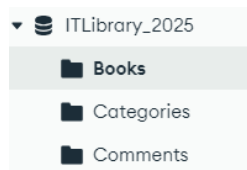
Data Structure:



Task 1: Database Setup

Create a database structure that supports book storage, categorization, and user comments:

- Using MongoDB Compass, create a database named **lab4**.
- Create collections named Books, Categories, and Comments based on the provided structure.
- Import sample data into these collections (*you can use the provided example data or create your own*).



Task 2: Backend API Development

In this question, you are asked to write an application, that provide some API to manage: Categories, Books and Comments. You are asked to build three APIs, as following:

GET	http://localhost:9999/books	Send
GET	http://localhost:9999/books/6603a6a7e9b5b3c7a1b8d020/comments	Send
POST	http://localhost:9999/books/create	Send

Task 2.1: Create an environment variables file

- Use a **.env** file to store the MongoDB connection string.

Task 2.2: The API at url: **http://localhost:3000/books**

The API at URL: **http://localhost:3000/books** returns information on all books in the database using the GET method, using GET method. Each book requires the information shown in Figure 1. Note that:

- **[comments]**: is a reference attribute of the Comments entities
- **[categories]**: is a reference attribute of the Categories entities

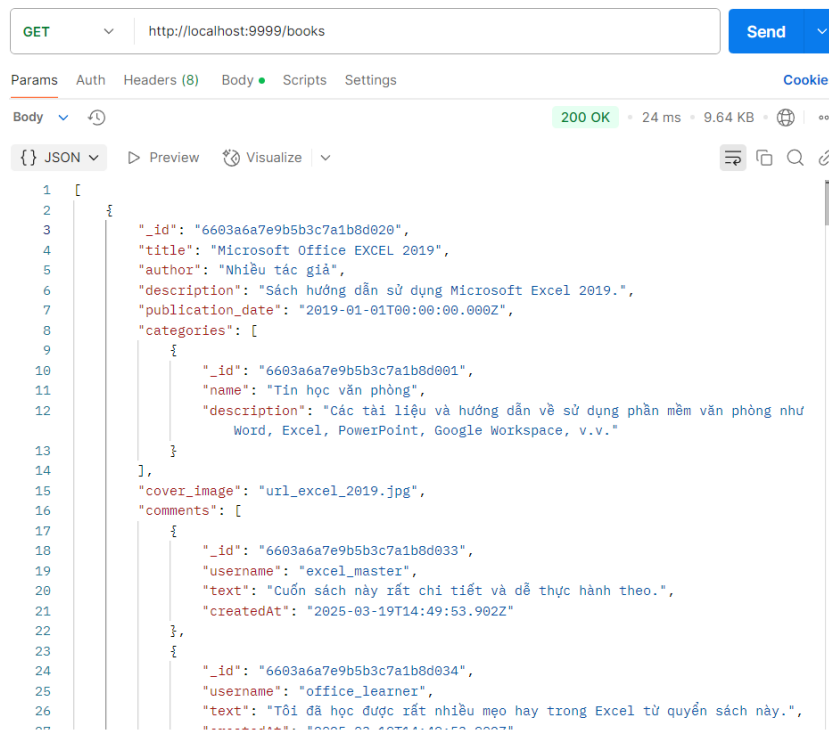


Figure 1 - The result of API at <http://localhost:3000/books> (using method GET)

- Reduce 0.5 points if the API does not display information about the Category.
- Reduce 0.5 points if the API does not display Comments.
- The Task will receive 0 points if the API does not function and fails testing in Postman.

Task 2.3: The API at url: <http://localhost:3000/books/:id/comments>

The API at URL: <http://localhost:3000/books/:id/comments> (:id is the ObjectId of a book) returns all comments for the given book ID in the database using the GET method. See Figure2 for more detail.

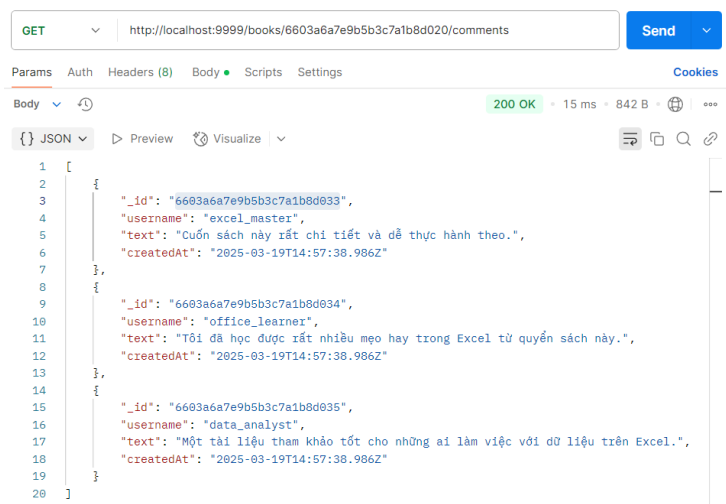


Figure 2 - The result of API at [/books/ 6603a6a7e9b5b3c7a1b8d020/comments](/books/6603a6a7e9b5b3c7a1b8d020/comments) using GET method

- Reduce 0.5 points if the API does not display the correct number of comments for each book.
- The Task will receive 0 points if the API does not display comments (except for books with no comments), displays incorrect information, or does not function and fails testing in Postman.

Task 2.4: The API at url: <http://localhost:3000/books/create>

The API at url: <http://localhost:3000/books/create>, using POST method to create a new Book. At the same time. See more detail.

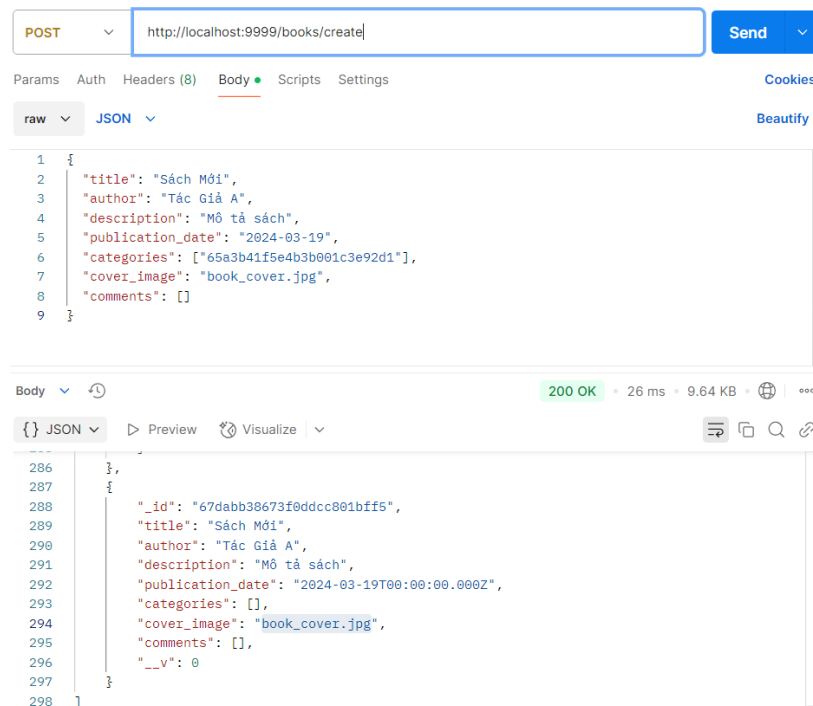
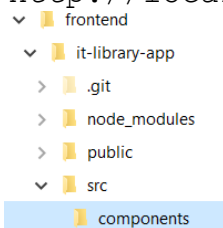


Figure 3 - Using POST method to create a new Book at url:
<http://localhost:3000/books/create>

- The Task will receive 0 points if the functionality does not work, or not functioning when tested in Postman.
- Reduce 0.5 points if the API does not insert all required fields into the Database when tested in Postman.

Task 3: Frontend Development using Created APIs

- In this question, you are asked to write a React web application (in front-end given folder), that using APIs of Task 2 to manage Books and Comments.
- Copy the **images** folder from the given download and paste it into the **public** folder in the project backend. If the **public** folder doesn't exist, create it in the root folder.
- Verify `REACT_APP_API_URL` in the **.env** file:
`REACT_APP_API_URL=http://localhost:3000`



Note: Only add code to handle the API call without modifying other parts of the code.

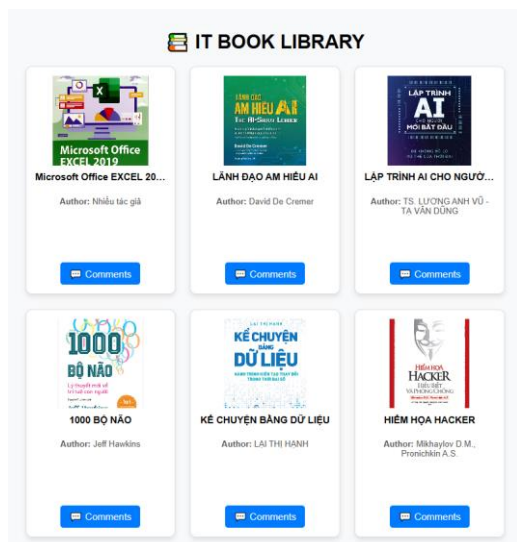


Figure 4 - The page when first loaded

Comments

User	Content	Created Date
excel_master	Cuốn sách này rất chi tiết và dễ thực hành theo.	3/19/2025, 10:23:00 PM
office_learner	Tôi đã học được rất nhiều mẹo hay trong Excel từ quyển sách này.	3/19/2025, 10:23:00 PM
data_analyst	Một tài liệu tham khảo tốt cho những ai làm việc với dữ liệu trên Excel.	3/19/2025, 10:23:00 PM

← Back

Figure 5 - List of comments by book ID at the URL:
<http://localhost:3000/books/6603a6a7e9b5b3c7a1b8d020/comments>

Task 3.1: Load all Books

When a user opens `http://localhost:3000/`, load all Books from the database like Figure 4.

Using the provided frontend project from the resources, install the necessary packages, then add code to the BookList component (*BookList.js* file) to display all books using the backend API. The *BookList.js* file has **3 areas** where API calls are required, code needs to be added.

Task 3.2: Display all comments related to the book

Using the provided frontend project from the resources, then add code to the CommentList component (*CommentList.js* file) to display all comments related to the book using the backend API like Figure 5.

. The *CommentList.js* file has **01 areas** where API calls are required, code needs to be added.